

PT1 Stance Modeling Update

ProtEvidenceDB: Abstract-Based Stance Pipeline Report

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This report summarizes the current abstract-level stance pipeline over cancer biomarker literature. The final setup no longer trusts the retrieval-time paper-to-pair link by itself. Instead, it validates gene and drug mentions directly inside each abstract, matches those mentions against the existing drug-gene interaction database, and only then computes the stance features and labels.

1. Intuition

1. Retrieval links are noisy. A paper found by a query does not always explicitly discuss the linked gene-drug pair.
2. Stance should be tied to local evidence, not the whole abstract. The pipeline therefore uses sentence-bounded matches and token windows.
3. Nearby phrases should count more than distant ones. Phase 3 uses a distance-weighted score: $\text{weight} = \text{base} / (1 + \text{distance_to_gene} + \text{distance_to_drug})$.
4. Generic neutral cues are downweighted so words like evaluated or review do not dominate the signal.
5. Evaluation is separated from raw extraction. Full phase outputs are written for all eligible papers; a separate 2,000-row CSV is produced for LLM/manual review.

2. Phase Definitions

Phase 1: Sentence-Level Bits. A paper receives binary indicators for whether a validated gene-drug pair appears in the same sentence as a positive, negative, or neutral phrase.

Phase 2: Sentence-Level Counts. This phase counts how many times those sentence-level co-occurrences appear. It is less strict than Phase 3 and is currently used as the sampling label for evaluation balancing.

Phase 3: Sentence-Bounded Token Windows. For each window size in {2, 5, 8, 10, 12, 16, 20}, the pipeline scores stance phrases by how close they are to both the gene and drug mention. Same-clause and between-entity phrase placements receive a modest boost.

Clarity / Unclear Logic. A Phase 3 label is marked as unclear when the strongest stance score is too weak or when the top two stance scores are too close to separate confidently.

3. Dataset Sizes

PubMed papers.csv rows: unavailable in current workspace snapshot

Papers with abstract-mentioned, interaction-validated gene-drug pairs: 19651

Evaluation sample size: 2000

The abstract-validated matching step removes many papers where the retrieval query found a paper, but the abstract itself does not state the relevant gene-drug pair.

4. Full-Dataset Distributions

Phase 1

Label	Count	Percent
supportive	1183	6.0%
opposing	444	2.3%
neutral	1366	7.0%
mixed	233	1.2%
unclear	16425	83.6%

Phase 2

Label	Count	Percent
supportive	1311	6.7%
opposing	465	2.4%
neutral	1420	7.2%
mixed	212	1.1%
unclear	16243	82.7%

Phase 3 (Window = 12)

Label	Count	Percent
supportive	158	0.8%
opposing	63	0.3%
neutral	1	0.0%
mixed	0	0.0%
unclear	19429	98.9%

5. Evaluation Sample Distributions

Sampling Label (balanced by Phase 2)

Label	Count	Percent
supportive	406	20.3%
opposing	400	20.0%
neutral	407	20.4%
mixed	212	10.6%
unclear	575	28.8%

Evaluation Labels: Phase 1

Label	Count	Percent
supportive	366	18.3%
opposing	381	19.1%
neutral	389	19.4%
mixed	221	11.1%
unclear	643	32.1%

Evaluation Labels: Phase 2

Label	Count	Percent
supportive	406	20.3%
opposing	400	20.0%
neutral	407	20.4%
mixed	212	10.6%
unclear	575	28.8%

6. Evaluation Labels: Phase 3 (Window = 12)

Label	Count	Percent
supportive	55	2.8%
opposing	59	3.0%
neutral	0	0.0%
mixed	0	0.0%
unclear	1886	94.3%

7. Interpretation

1. Phase 1 and Phase 2 show materially more non-unclear labels after switching to abstract-validated gene-drug pairs.
2. Phase 3 remains much more conservative because it requires local proximity and confidence separation.
3. The balanced evaluation sample is driven by Phase 2 because Phase 3 alone does not yet produce enough non-unclear cases for a useful manual review set.
4. This setup is suitable for the next stage: LLM or manual adjudication over the 2,000-row evaluation sheet, followed by supervised classifier training.